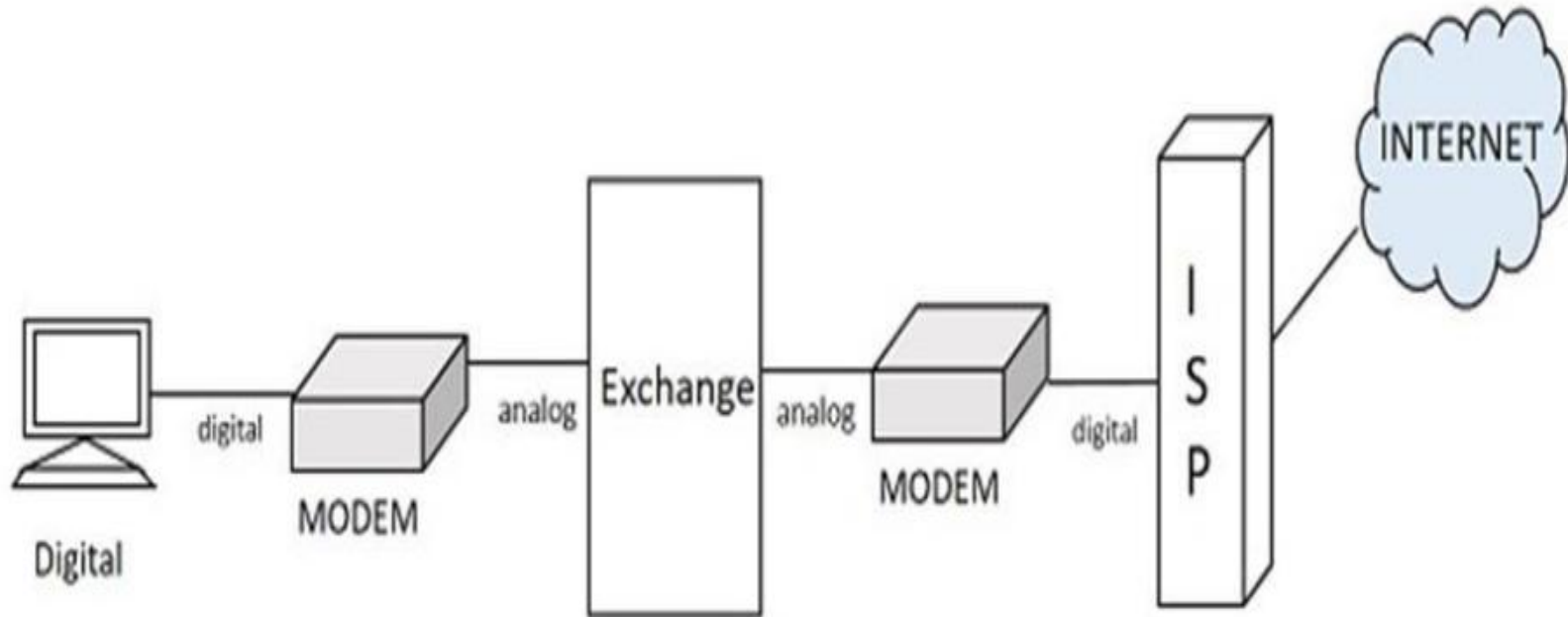


INTEGRATED SERVICES DIGITAL NETWORKS

integrated services digital networks:

- The transmission of data and voice both were possible through normal POTS, Plain Old Telephone Systems.
- The process of connecting a home computer to the Internet Service Provider used to take a lot of effort.
- The usage of the modulator-demodulator unit, simply called the MODEM was the essential thing to establish a connection.

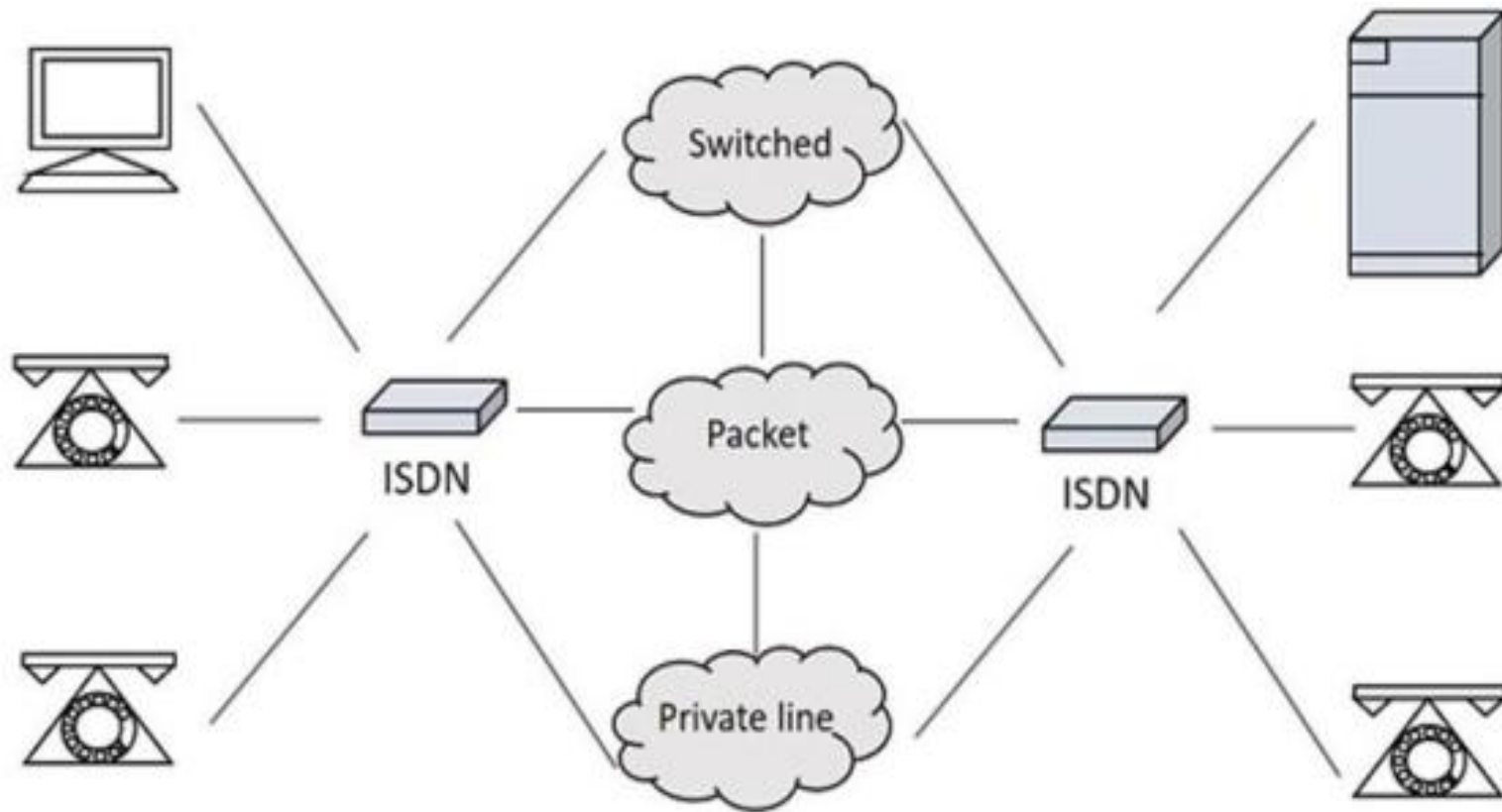
integrated services digital networks:



ISDN

- The introduction of ISDN has resolved this problem allowing the transmission of both voice and data simultaneously.
- This has many advanced features over the traditional PSTN, Public Switched Telephone Network .
- ISDN was first defined in the CCITT red book in 1988. The Integrated Services of Digital.
- Networking, in short ISDN is a telephone network based infrastructure that allows the transmission of voice and data simultaneously at a high speed with greater efficiency.

ISDN



ISDN supports a variety of services. A few of them are listed below:

- Voice calls
- Facsimile
- Videotext
- Tele text
- Electronic Mail
- Database access
- Data transmission and voice
- Connection to internet
- Electronic Fund transfer
- Image and graphics exchange
- Document storage and transfer
- Audio and Video Conferencing
- Automatic alarm services to fire stations, police, medical etc

Types of ISDN

- Among the types of several interfaces present, some of them contains channels such as the B-Channels or Bearer Channels that are used to transmit voice and data simultaneously; the D Channels or Delta Channels that are used for signalling purpose to set up communication.
- The ISDN has several kinds of access interfaces such as:
 - Basic Rate Interface (BRI)
 - Primary Rate Interface (PRI)
 - Narrowband ISDN
 - Broadband ISDN
- Broadband Communications. The broadband services provide a continuous flow of information, which is distributed from a central source to an unlimited number of authorized receivers connected to the network.

Advantages of ISDN

- ISDN is a telephone network based infrastructure, which enables the transmission of both voice and data simultaneously.

There are many advantages of ISDN such as:

- As the services are digital, there is less chance for errors.
- The connection is faster.
- The bandwidth is higher.
- Voice, data and video – all of these can be sent over a single ISDN line.

Disadvantages of ISDN

- The disadvantage of ISDN is that it requires specialized digital services and is costlier.
- However, the advent of ISDN has brought great advancement in communications. Multiple.
- transmissions with greater speed are being achieved with higher levels of accuracy .

Motivation for ISD

- Three factors are responsible for the developments towards ISDN:
 1. Sociological or societal needs
 2. Economic necessities
 3. Technological developments.

ISDN supports a variety of services. A short list of some of the important services is:

- 1 Videotex
2. Electronic mail
3. Digital facsimile
4. Teletex
5. Database access.

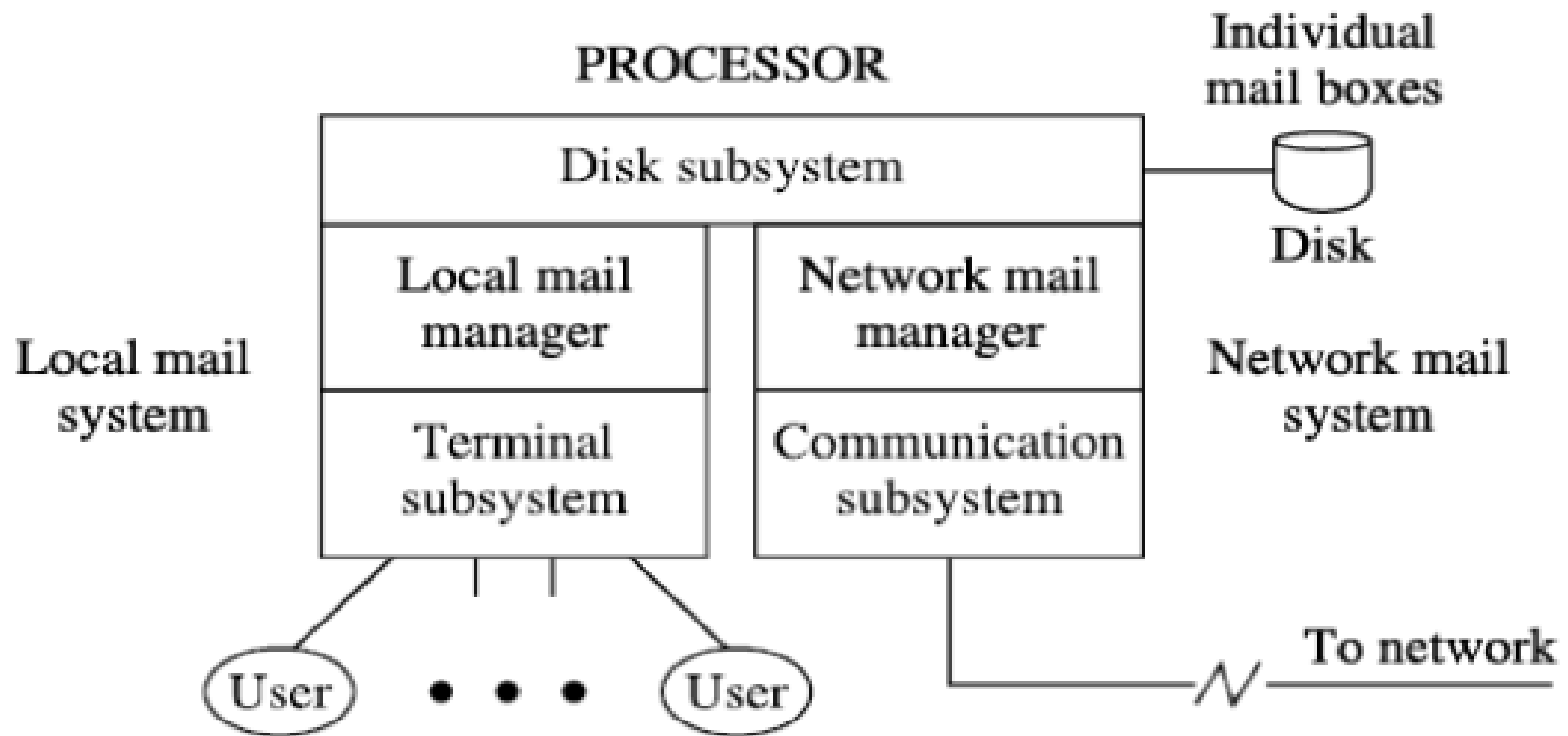
These services are described in the following sections.

Videotex

Video tex is a generic term for systems that provide easy to use. Low cost computer based Services via communication facilities. Three forms of video texexist:

1. View
2. Tele text
3. Open channel teletext.

Electronic Mail

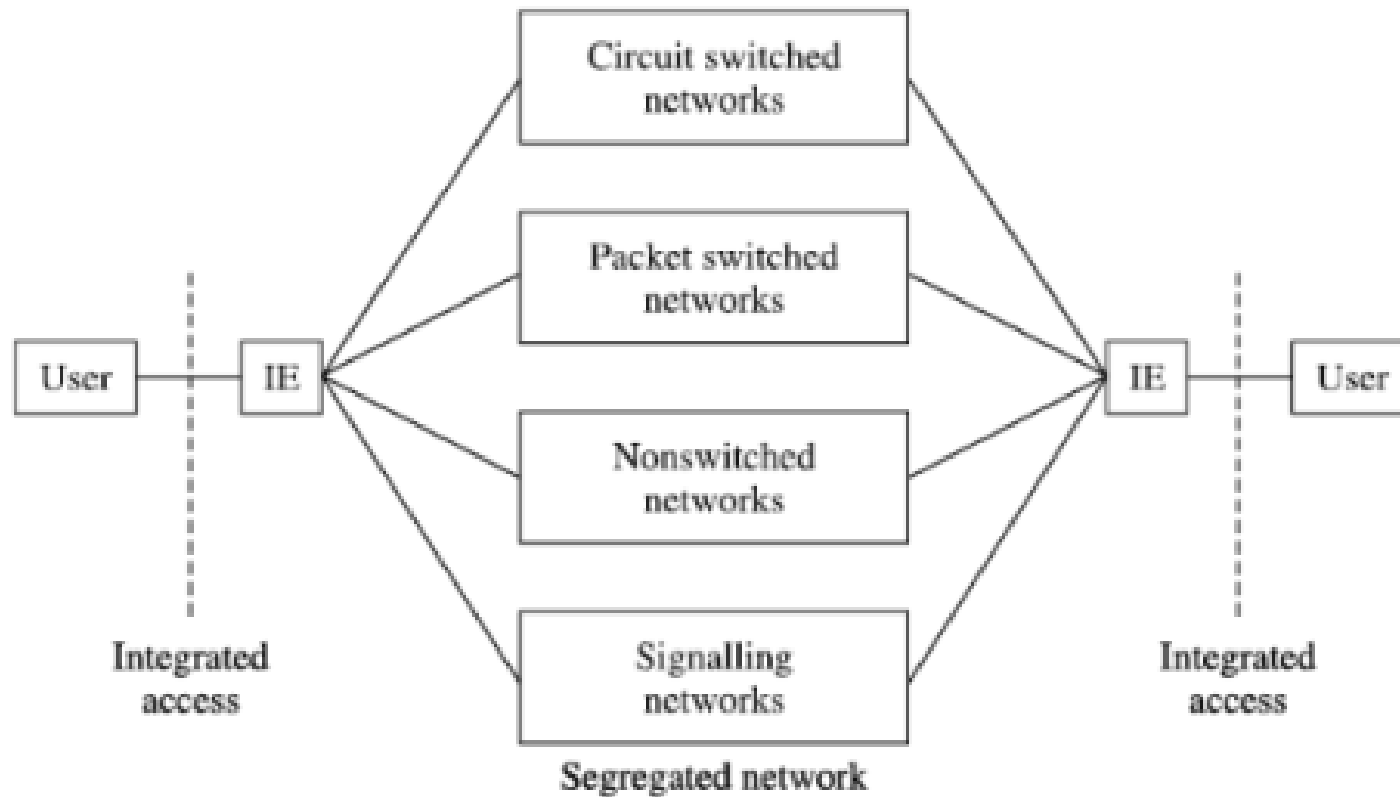


A Typical configuration of the an electronic mail system

NETWORK AND PROTOCOL ARCHITECTURE

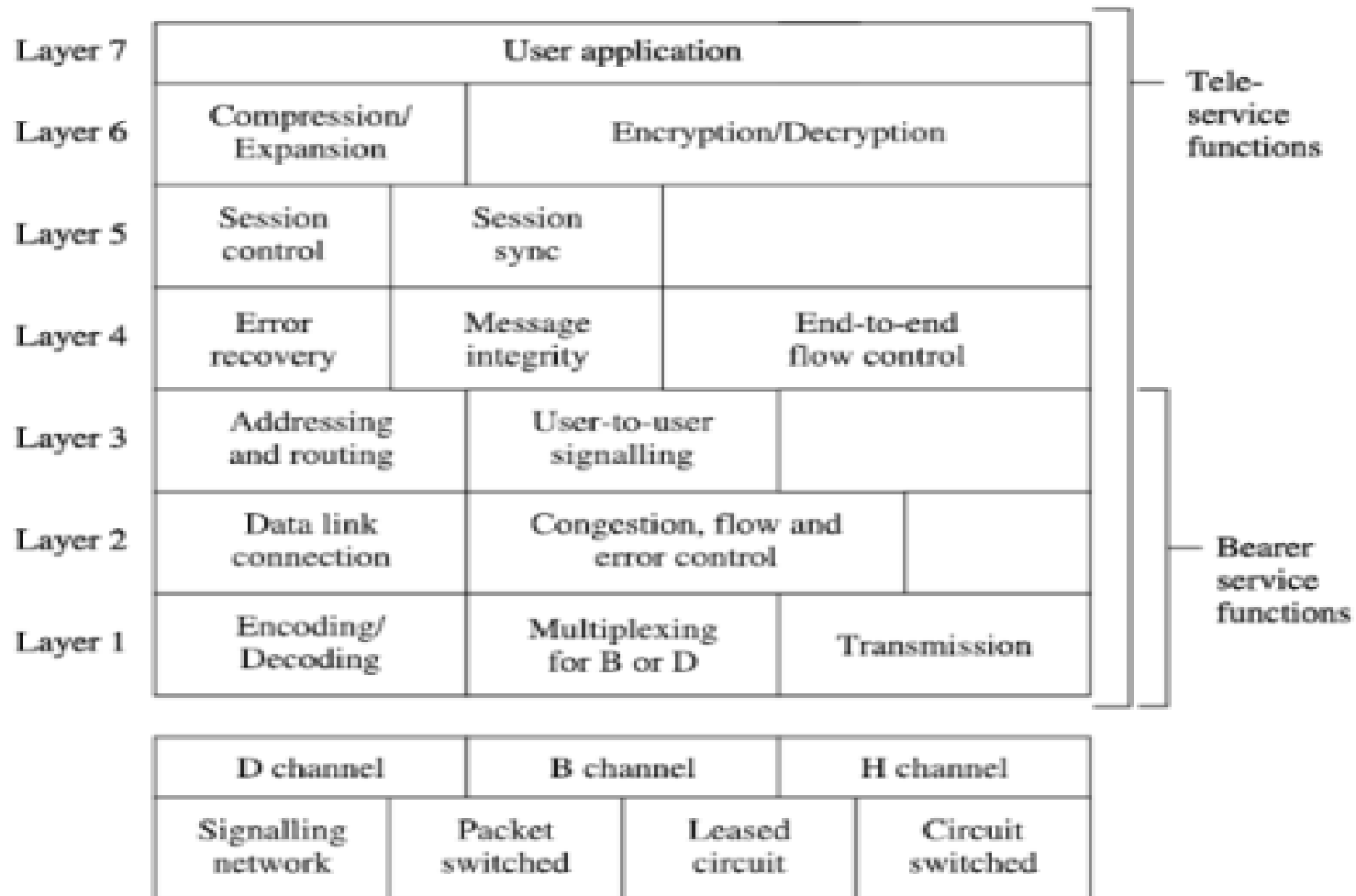
- Network architecture of [SDN followed an evolutionary path.
It is natural that an evolutionary approach was taken with regard to ISDN.
- The wide range of telecommunication equipments and networks that were existing could not be replaced overnight by ISDN.
- As a first step, the then existing analog telephone networks were converted to digital networks.
- These networks were then to be operated along with other existing data and signalling networks.
- It is seen from the figure that four different types of networks form part of the segregated architecture.
 - An ISDN exchange and a suitable user-network interface permit an integrated access to the network facilities.

ISDN Architecture



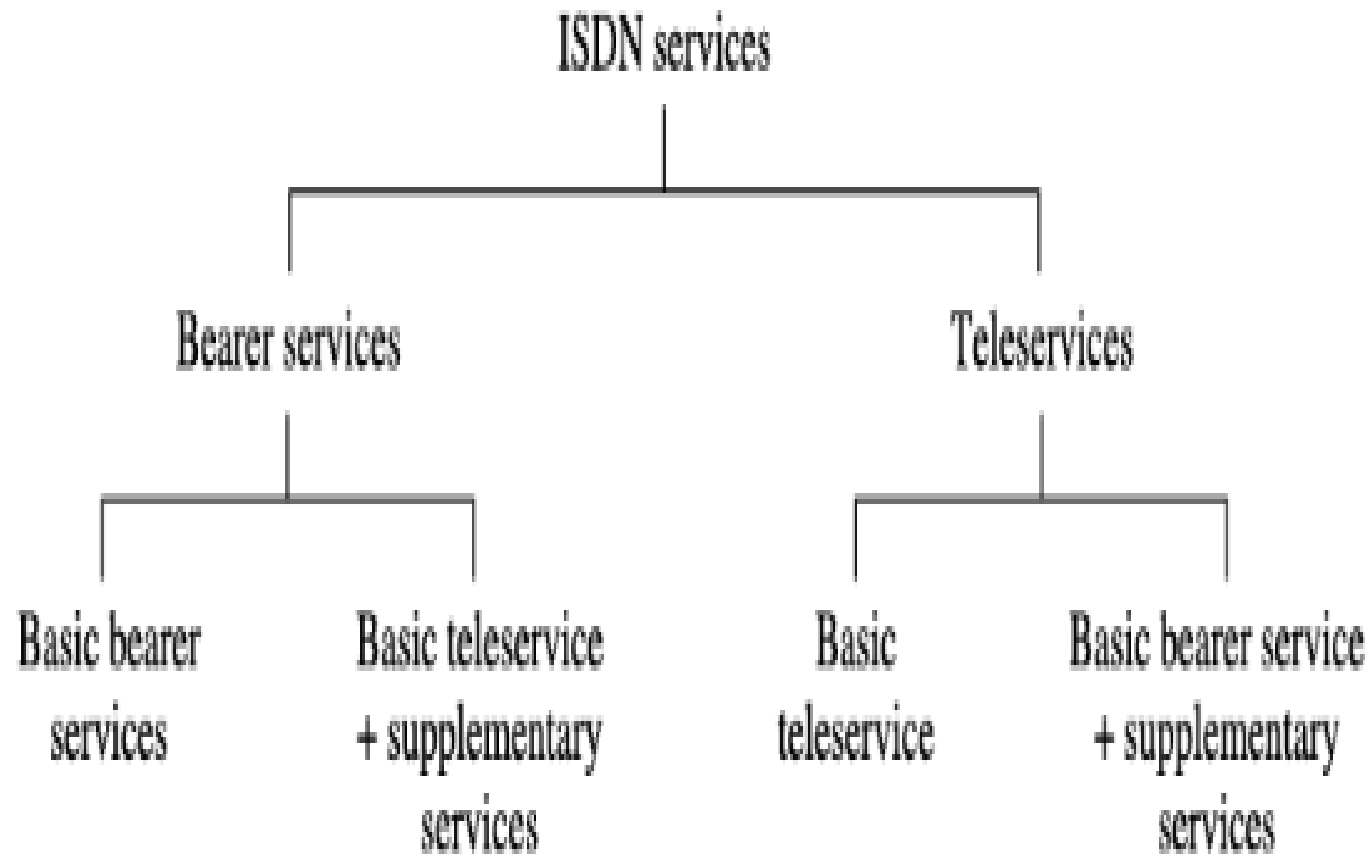
ISDN Architecture

ISDN Protocol architecture

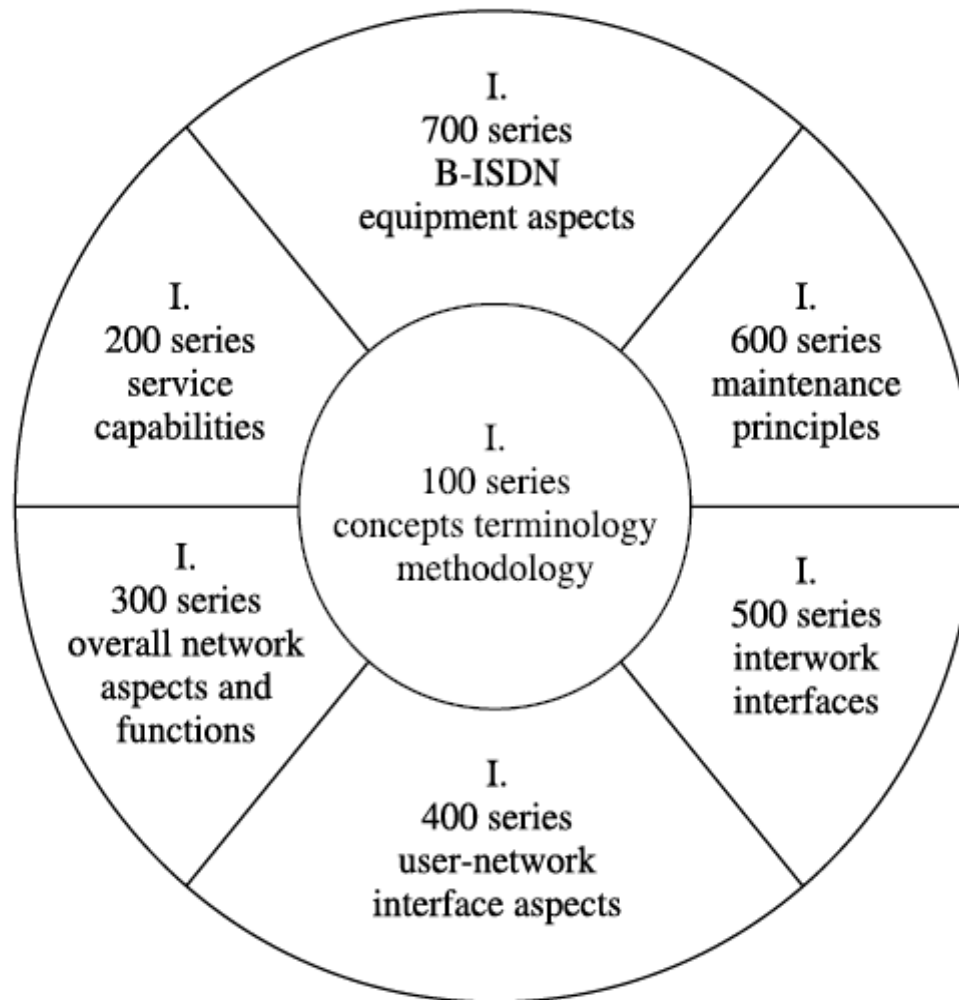


ISDN Protocol architecture

ISDN Services



ISDN STANDARDS



BROADBAND ISDN

- Broadband ISDN (BISDN) is defined as a network capable of supporting data rates greater than the primary rate (1S44 or 2.048 Mbps) supported by ISDN. In the context of BISDN,
- The original ISDN concept is often termed narrowband ISDN
- (NISDN). BISDN services are broadly classified as
 - . Interactive services
 - . Distribution services.

Interactive services may be classified as

1. Conversational sen-ices
2. Messaging services
3. Retrieval services.

Distribution services are classified as

1. Broadcast services
2. Cyclic services.

VOICE DATA INTEGRATION

Digitised voice	Data
Periodic bursty in nature	Aperiodic bursty in nature
Fixed length bursts	Variable length bursts
Small packet size	Large packet size
Packetisation time critical	Packetisation time is not critical
Hard bound on delay	Soft bound on delay
Hard bound on the variance of delay	Soft bound on the variance of delay
Loss of parts of speech acceptable	Loss of parts of data unacceptable
Low overhead as there is no error recovery	High overhead due to error detection and recovery